

Evaluating a Bait Capture Enrichment Approach for the Zebrafish Chemical Defensome

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Studying the Chemical Defensome of Zebrafish

Research Objective: Understand how chemical contaminants are metabolized in Zebrafish through transcriptome analysis

Method:

- Assess transcriptomic responses of male and female zebrafish:
 - In different tissues
 - To different contaminants
 - At different time points

Complications:

- RNA-sequencing is expensive
 - $2 \text{ tissues} \times 2 \text{ conditions} \times 4 \text{ timepoints} \times 2 \text{ sexes} = 32 \text{ transcriptomes}$
- Bait-capture enrichment allows us to examine specific genes at a reduced cost
 - Will it actually provide an accurate estimate of transcript counts?

The Data and our Question

RNA was extracted from male and female zebrafish liver tissue for:

Conventional RNA-sequencing:

- Measures count for every transcript
- Requires large sequencing depth (~30M)
- \$150-\$300 per sample

Bait-Capture Enrichment RNA-Sequencing:

- Measures counts for specific transcripts (595 targets)
- Requires much lower sequencing depth (~6M)
- \$30-\$60 per sample

Our Main Question:

Does the bait-capture enrichment protocol distort transcript abundances?

General Principles of Equivalence Tests

Null hypothesis tests are used to provide evidence of a clear effect

- Are these two values clearly different?

Equivalence testing are used to provide evidence of equivalence

- Are these two values *practically* equivalent?
- Not that there is *no effect*

Complication: a reversed H_0 of “not zero” is near impossible to reject

- Normal variability will always lead to some small effect
- Theoretically, any treatment has an effect, even if it is tiny

Defining Equivalence

Equivalence cannot mean *identical*

- Are two values equivalent, within bounds I consider negligible?

The user must define 'equivalence'

- Choosing and justifying equivalence bounds is difficult
- Select based on:
 - Prior evidence
 - Practical importance
 - Expert opinion

Choice of equivalence bounds
is always somewhat arbitrary

Testing for Equivalence: One-Sample TOSTs

Two one-sided T-tests (TOST):

$$t_L = \frac{M - \mu - \Delta_L}{\frac{SD}{\sqrt{N}}} \text{ and } t_U = \frac{M - \mu - \Delta_U}{\frac{SD}{\sqrt{N}}},$$

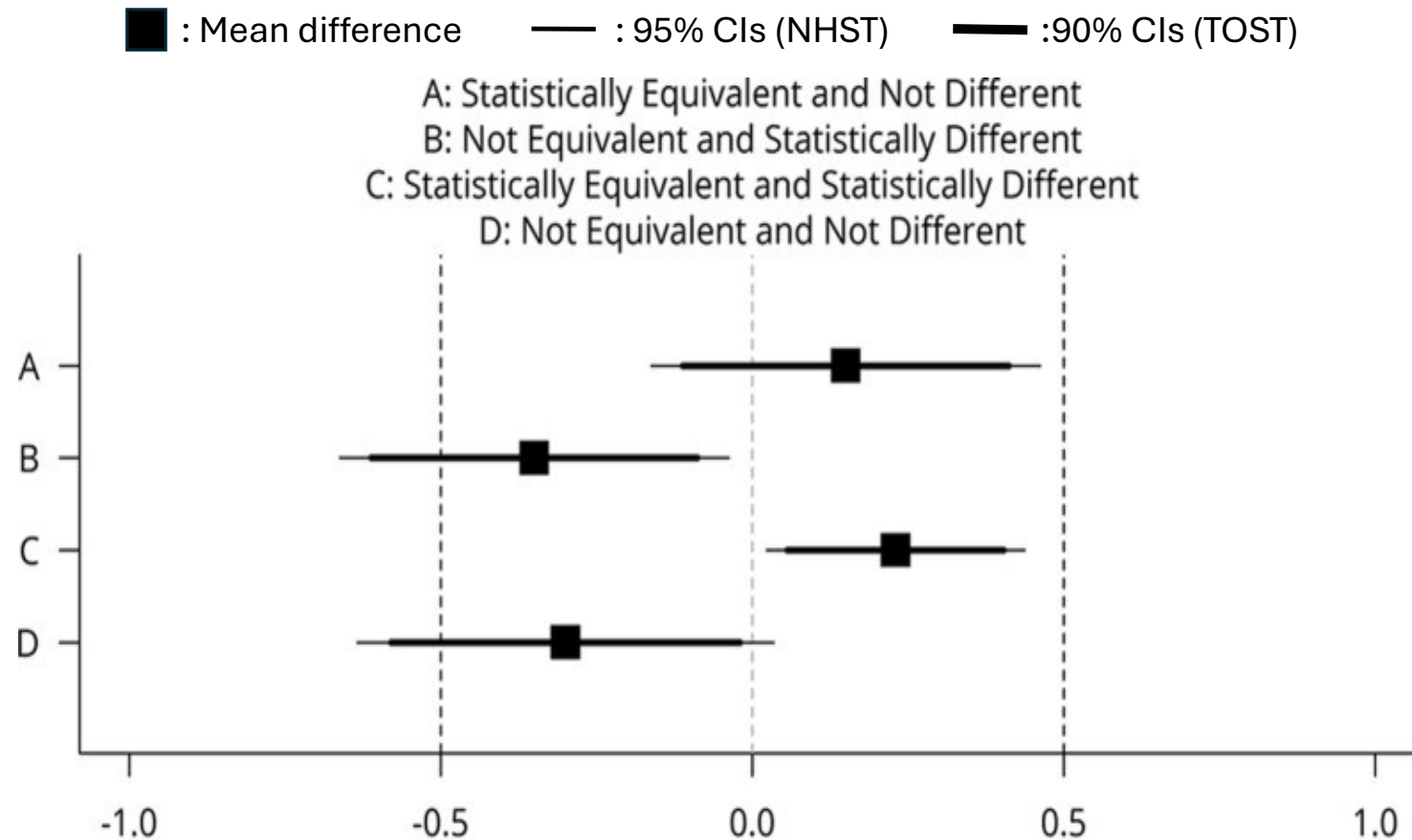
M → Mean
μ → Value to test against
Δ_{U/L} → Equivalence Bounds
SD → Standard Dev.
N → Sample Size

$$H_0 : \Delta_L \geq M - \mu \text{ or } M - \mu \geq \Delta_U$$

$$H_1 : \Delta_L < M - \mu < \Delta_U$$

i.e., is the difference clearly less than or greater than my equivalence thresholds?

Testing for Equivalence: TOSTs



Using TOSTs to test equivalence

Objective: Are the enriched and unenriched data *practically equivalent*?

Step 1: Normalize counts for BC and conventional RNA-seq data

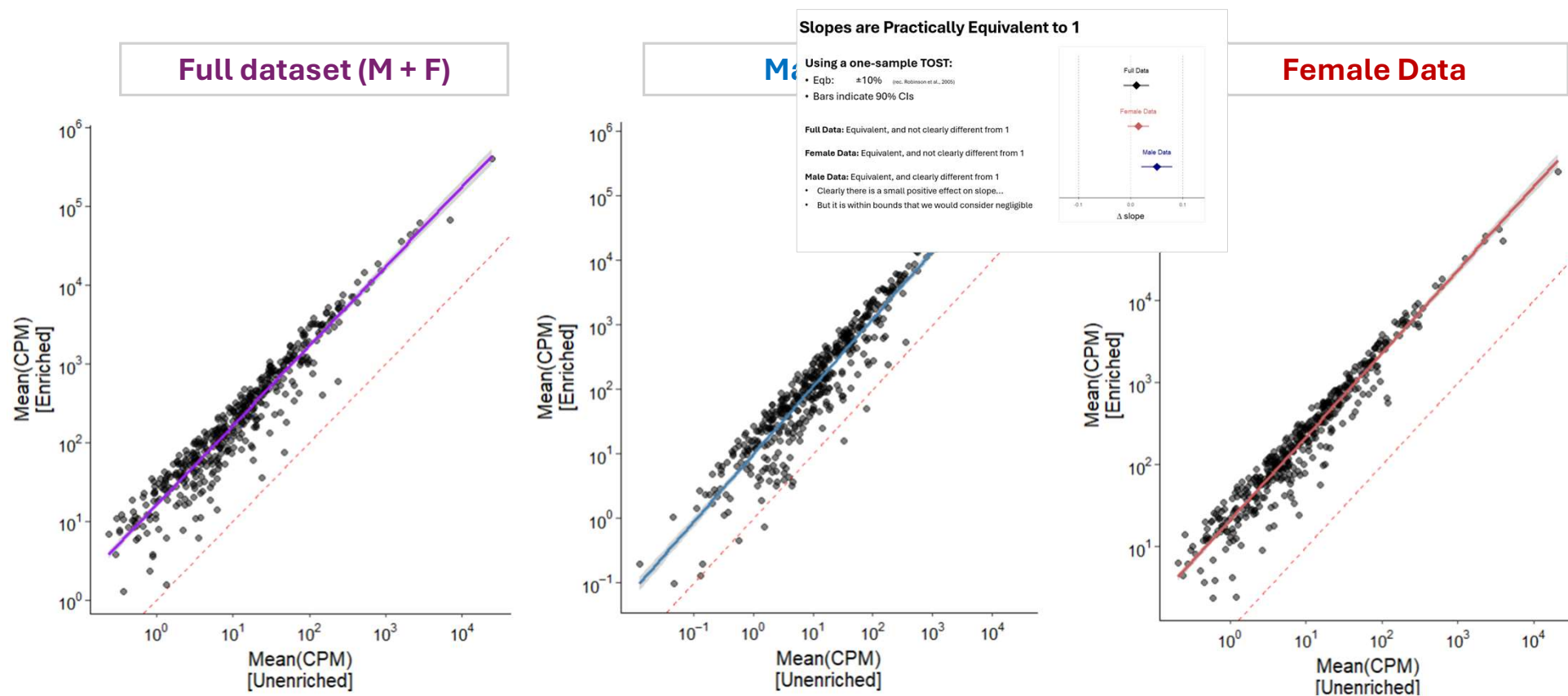
Step 2: Fit a linear model and test the slope for ~equivalence to 1

The general idea is:

If the bait-capture enrichment ***distorts*** counts: Slope $\neq 1$

If the bait-capture enrichment ***maintains*** counts: Slope ~ 1

Linear Models: Total, Male, and Female Data



Q: Is the slope of the linear model practically equivalent to one?

Slopes are Practically Equivalent to 1

Using a one-sample TOST:

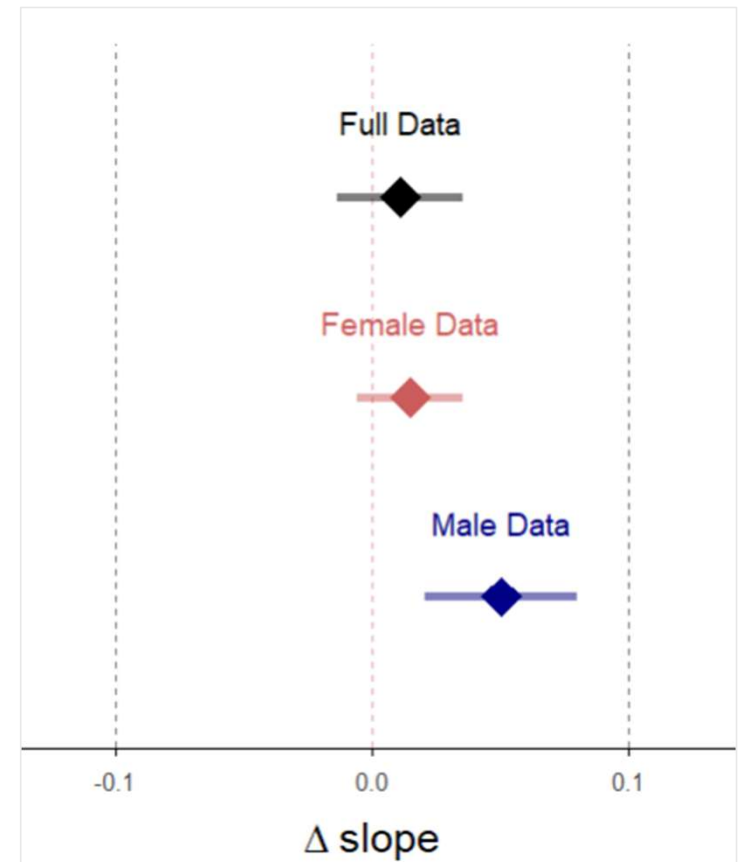
- Eqb: $\pm 10\%$ (rec. Robinson et al., 2005)
- Bars indicate 90% CIs

Full Data: Equivalent, and not clearly different from 1

Female Data: Equivalent, and not clearly different from 1

Male Data: Equivalent, and clearly different from 1

- Clearly there is a small positive effect on slope...
- But it is within bounds that we would consider negligible



Are Individual Genes Equivalent?

Globally, Bait-Capture Enrichment does not appear to distort counts...

- What about counts for individual genes?
- Are the counts for any individual gene distorted by BC enrichment?

Approach:

Test for equivalence using relative expression levels

- Differences in raw counts and sequencing depth made direct comparison impossible
- Relative expression should allow for direct comparison of genes

Complication: No reference gene was included...

Relative Expression: BC vs. Conventional RNA-seq

Choice of a Reference Gene:

- No conventional "housekeeping genes" were included in the probe targets
- We needed to identify the most *stable gene*

geNorm Ranking:

- Compares each gene with every other gene in each sample and compute total variation
- Takes the geometric mean of these values to use as an "M-score"
- Removes the gene with the highest M-score (i.e., the least stable)
- Repeats until only 2 genes remain (most stable)

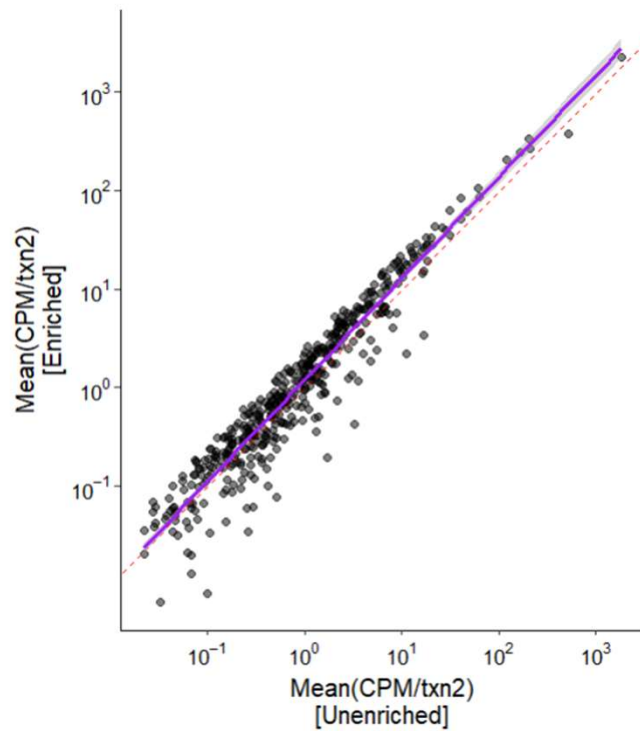
Gene_ID	M_en	Rank_en	M_un	Rank_un	Combined_rank
<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
txn1l	0.352	11	0.379	11	22
clpp	0.337	10	0.367	10	20
slc30a5	0.324	9	0.290	5	14
abcg2a	0.307	8	0.354	9	17
dnajb1a	0.296	7	0.341	8	15
nat9	0.283	6	0.309	6	12
slc22a15	0.275	5	0.118	1	6
ugt2a3	0.266	4	0.237	3	7
ephx5	0.241	3	0.118	1	4
dnajc4	0.210	2	0.326	7	9
akr1a1b	0.101	1	0.271	4	5
txn2	0.101	1	0.199	2	3

txn2 is the most stable
choice for a reference

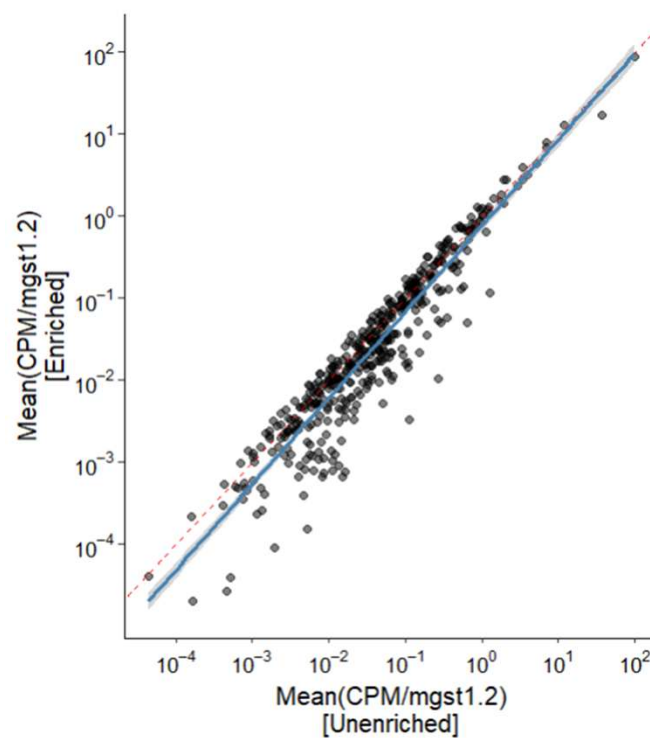
Relative Expression: Enriched vs. Unenriched

Intercepts are now close to zero

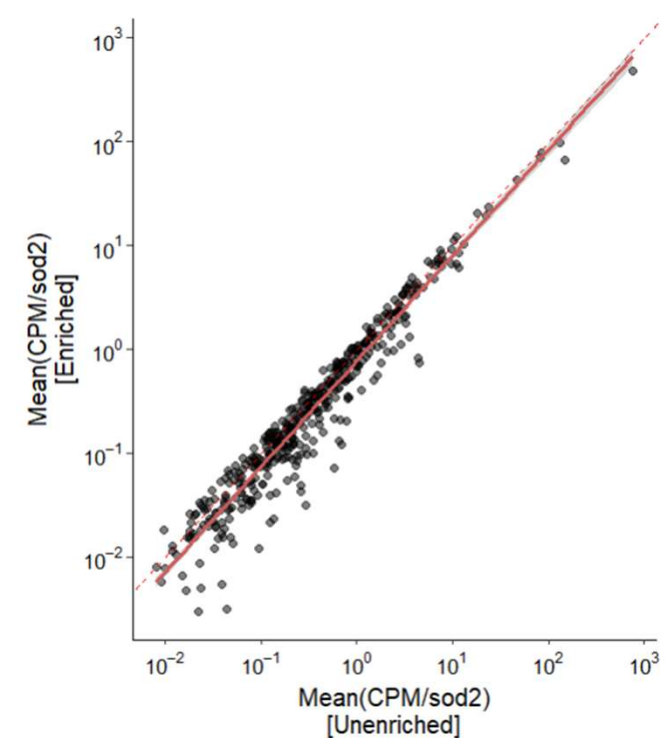
Full dataset (M + F)



Male Data



Female Data



i.e., Relative expression in enriched/unenriched are on a comparable scale

Testing for Equivalence: TOSTs

Two one-sided T-tests (TOST):

Testing if two means are ~equivalent...

$$t_L = \frac{\bar{M}_1 - \bar{M}_2 - \Delta_L}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \text{ and } t_U = \frac{\bar{M}_1 - \bar{M}_2 - \Delta_U}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}, \quad \sigma = \sqrt{\frac{(n_1 - 1) SD_1^2 + (n_2 - 1) SD_2^2}{n_1 + n_2 - 2}}.$$

$$H_0 : \Delta_L \leq \bar{M}_1 - \bar{M}_2 \quad \text{or} \quad \bar{M}_1 - \bar{M}_2 \geq \Delta_U$$

$$H_1 : \Delta_L < \bar{M}_1 - \bar{M}_2 < \Delta_U$$

i.e., is the difference in means clearly less than or greater than my equivalence thresholds?

Using TOST to Test ~Equivalence

Objective: Is the relative expression of genes *practically equivalent*?

Step 1: Calculate relative expression for each gene to a stable reference

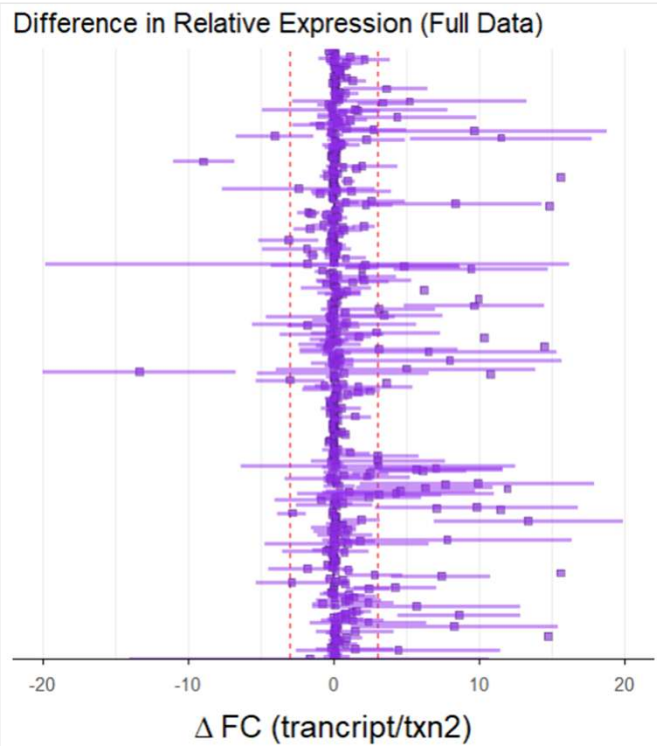
Step 2: Perform TOSTs to test if the mean relative expression is equivalent

Complications:

- Since there are sex differences, looking at the total dataset in this case is probably not very useful
- Multiple test corrections (CIs?)

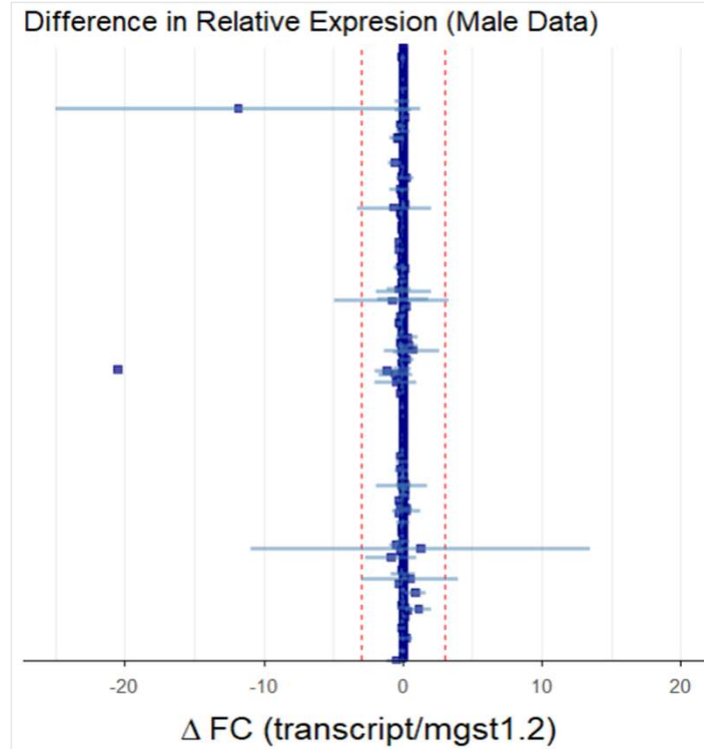
TOSTs for equivalence of individual genes

Full dataset (M + F)



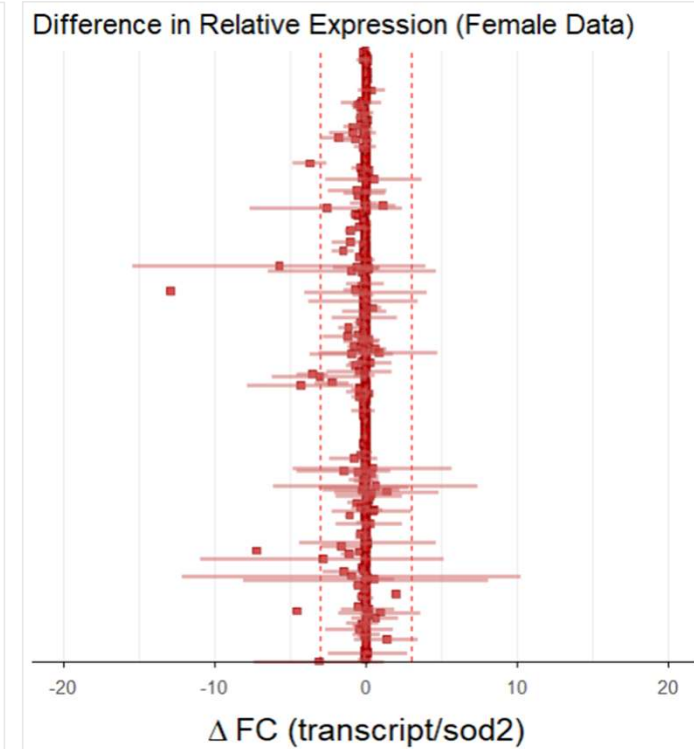
325/437

Male Data



429/436

Female Data



403/437

Relative Expression for most genes is ~Equivalent

Total Data:

- Never expected TOST to show most genes as ~equivalent
- Sex differences in expression would inflate variation

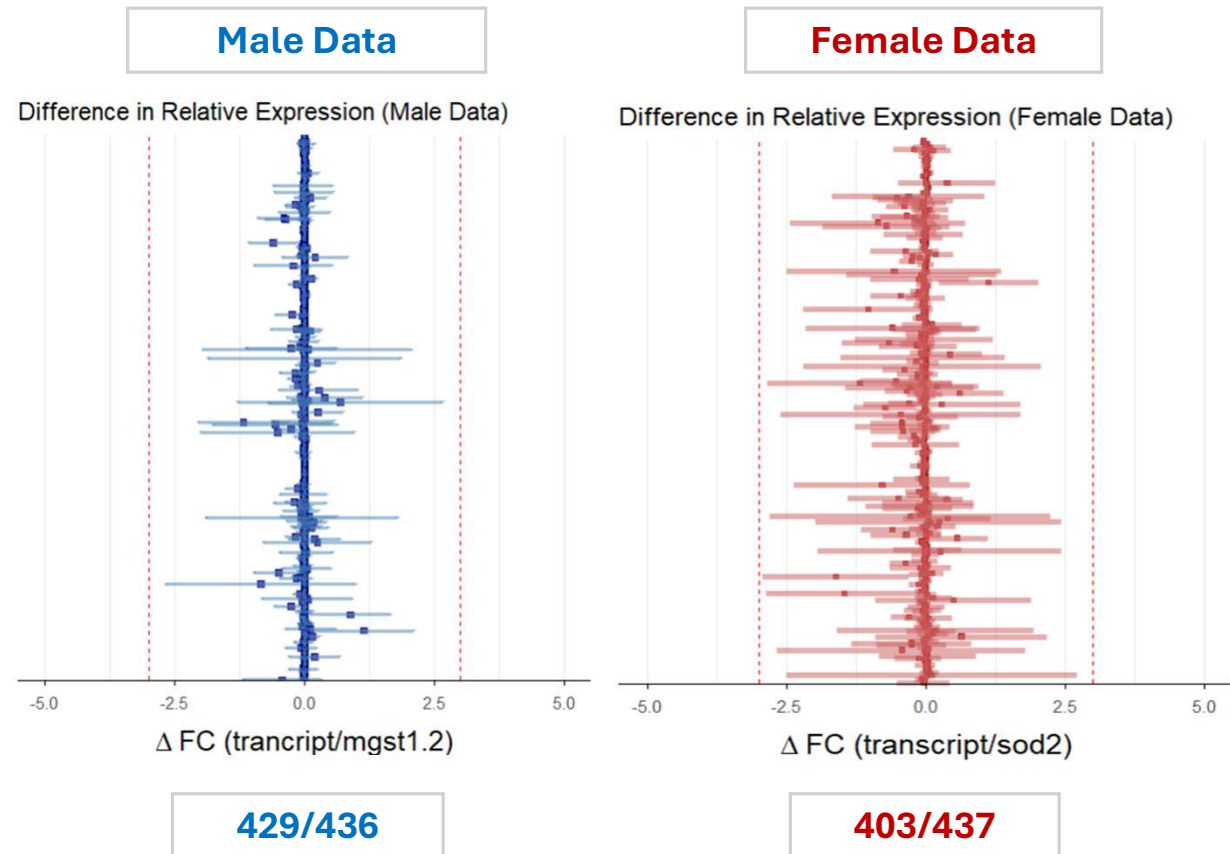
Male/Female Data:

- *Most* genes are ~equivalent

New Questions

Which genes are not ~equivalent?

Are the equivalence bounds I chose ($\Delta FC = 3$), too conservative?



What's Next?

Looking closer at individual genes:

- Interrogate "non-equivalent genes":
 - How different are they?
 - Is it bad enough to affect biological interpretations?
 - Is there an obvious reason for the differences (i.e., probe chemistry)?

Compare differential gene expression analysis results

- Compare enriched Male vs. Female, and unenriched Male vs. Female
- DEGs should be ~equivalent

Applying our TOST results to differential gene expression analysis

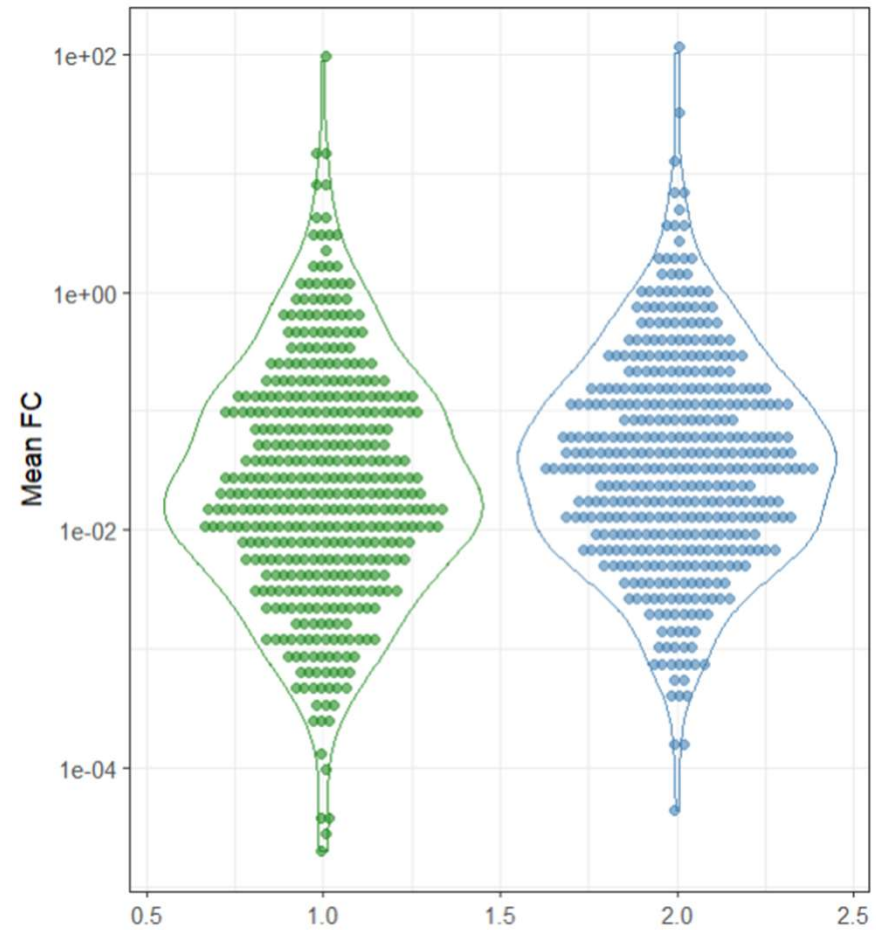
- Compare enriched liver data to other tissues
- Explore how our "non-equivalent" genes are being represented

Extra Stuff: Is the magnitude of FCs different?

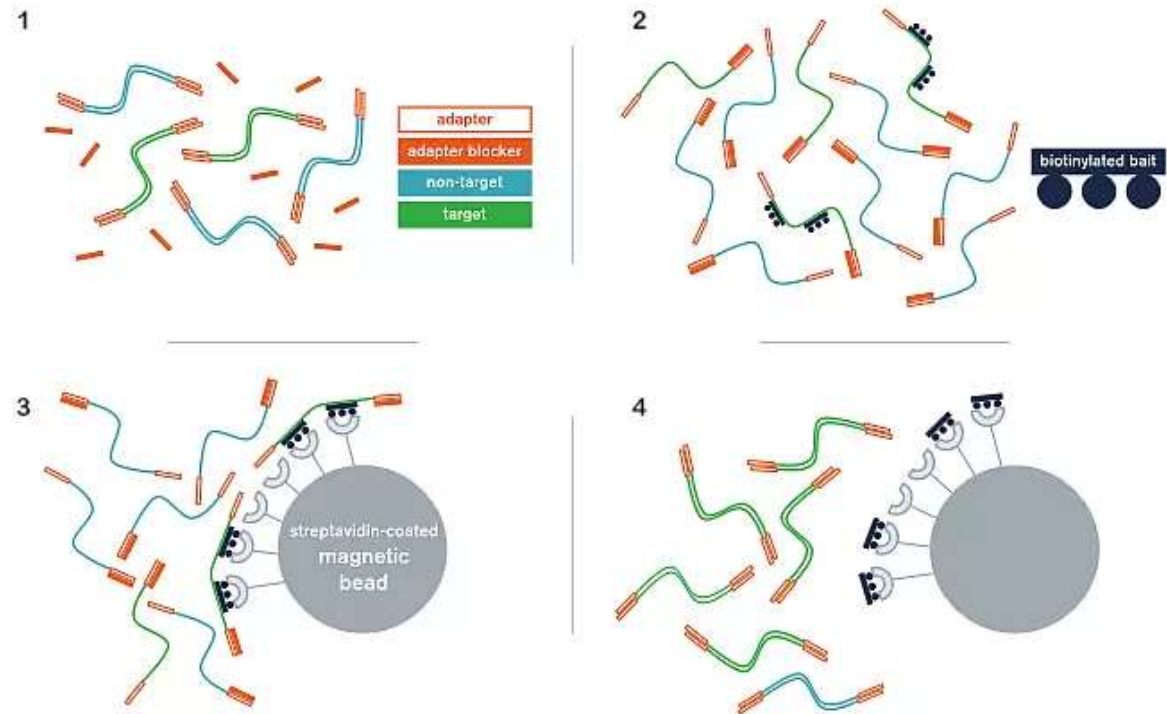
Not really;

Left is enriched; right is
unenriched

Plotted mean relative
expression for both for
male data; pretty similar



1. Adaptor Ligation + cDNA synthesis
2. Probe hybridization to complementary sequences
3. Biotinylated bases bind to magnetic beads
4. Wash sample to separate target cDNA



What is Baits Enrichment

- Targeted sequencing method used to isolate specific DNA/RNA regions of interest
- Typically used to enrich low-abundance targets from complex sample
- Better resolution of read counts for lowly expressed genes at lower sequencing depths to help lower sequencing costs
- Allows for better understanding of specific processes involving lowly expressed genes but doesn't allow for global expression patterns